Integers, Primes & Powers ⏱ ~25 min

NAME _____ DUE _____ SCORE / EFFORT _____

No calculator. Show your working.

1 · This week

1. Evaluate.

a) $-15-(-7)+3$ _____

b) $(-4)^2-4^2$ _____

c) $48\div(-6)\times(-2)$ _____

d) $-3\times[4-(-2)]^2$ _____

e) $(-1)^{2027}+(-1)^{2026}$ _____

WORKING

2. Write as a product of primes in index form.

a) 252 _____ b) 945 _____ c) 1024 _____

WORKING

3. Use your answers to Q2 to find the HCF and LCM of 252 and 945. Check using HCF × LCM = product of the numbers.

WORKING

4. Lighthouses. Three lights flash together. Light A flashes every 15 seconds, Light B every 20 seconds and Light C every 25 seconds. How long until all three flash together again? Give your answer in minutes.

WORKING

5. Find the smallest positive whole number that leaves a remainder of 1 when divided by 2, 3, 4, 5 **or** 6, but is exactly divisible by 7.

WORKING

2 · Spaced revision

6.

a) $\frac{2}{5}$ of $\frac{3}{4}$ _____

b) $1\frac{3}{4} - \frac{5}{6}$ _____

c) Increase $80 by 15%. _____

d) Write 0.375 as a fraction in simplest form. _____

WORKING

3 · Think deeper / At home

7. Explain to someone at home **why** a negative times a negative is positive. Use this pattern and continue it for three more lines:

$-3 \times 2 = -6$

$-3 \times 1 = -3$

$-3 \times 0 = 0$

$-3 \times -1 = $ _____

…

What do you notice about how the answers change each line? Write one sentence explaining what your listener said or asked.